5-Minute Notebook Walkthrough (Compact) + Q&A Deep-Dive Reference

Target File to Scroll: `notebooks/commodity_lstm_trading_v2.ipynb` (or `.html`)
Total Time Budget: 5:00 minutes (**3:15 Gabriel** [Sections 1–6, `Out [1]-[8]`] | **1:45 Tomàs** [Sections 7–8, `Out [9]-[11]`])

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1. Live 5-Minute Walkthrough (Core Talking Points Only)

Part 1 — Gabriel (`0:00 - 3:15` | 3 min 15 sec — Sections 1 to 6)

Title & Section 1: Data & Stylized Facts (`0:00 - 0:45` | 45 sec) — *Scroll to `Out [1]` & `Out [2]`*

- **Goal:** Test whether a small neural network with memory [**compact LSTM**] beats simple trend rules [**Simple Momentum**] and standard ML [**RidgeCV, LightGBM**] after trading costs [**Net Sharpe Ratio**] across 5 commodity ETFs: broad commodities, oil, gold, silver, and agriculture [**DBC, USO, GLD, SLV, DBA**].
- **Clean Setup:** Aligned from Jan 2007 (`5,158` business days) [**synchronized inception**], predicting tomorrow's return from today's data [**zero look-ahead bias,** $y_t = r_{t+1}$], scaled to 10% yearly risk [**`TARGET_VOL` covariance scaling**].
- **Why Commodities Need This (`Out [1]` & `Out [2]`):**
  1. Oil (`37.5%` vol) and silver (`33.0%`) swing twice as hard as agriculture (`16.7%`) and gold (`18.1%`) [**volatility dispersion**] → requires risk scaling.
  2. Passive oil (`USO`) earns `0.07` Sharpe due to monthly futures rolling losses [**contango / negative roll yield**], whereas active trend trading can short downturns and ride rallies [**backwardation**].
  3. Frequent extreme crash days [**fat tails / kurtosis**] motivate a crash-resistant loss [**Huber loss**], while low gold–oil correlation (`0.20`) gives strong diversification.

Section 2 & 3: Features & LSTM Architecture (`0:45 - 1:35` | 50 sec) — *Scroll to `Out [3]` & `Out [4]`*

- **Multi-Speed Trend Inputs:** Instead of just 1-day returns [`X_raw`] or 1-year trends [`X_mom_252`], we stack 1-week, 1-month, 3-month, 6-month, and 1-year average returns (`5d-252d`) across the 5 ETFs [**multi-horizon momentum `X_mom`, 25 features**].
- **Walk-Forward Training:** Retrained monthly (`21d`) on a rolling 3-year window (`756d`), tested out-of-sample on **2020–2026** [`skfin.Backtester`].
- **Anti-Overfitting Design (`Out [4]`):** Because daily returns are mostly noise [**stationary returns**], we use a tiny 8-unit network [`hidden_size=8`], 20% dropout, strong weight penalty [`weight_decay=1e-3`], crash-resistant **Huber loss**, chronological early stopping on the newest 15% of training data, and a 20-day trailing buffer [`last_train_X_`] so test folds have full history.

Section 4 & 5: Pre-2020 Calibration vs. 2020–2026 Shocks (`1:35 - 2:25` | 50 sec) — *Scroll to `Out [5]` & `Out [6]`*

- **Fair Calibration (`2012-2019`, `Out [5]`):** A 15-trial random search before the test period favored a longer 32-day memory with weak regularization [**Trial 9: `lookback=32, hs=9, wd=3.4e-4`, IS Sharpe `0.65`**] because 2012–2019 was calm.
- **Out-of-Sample Results (`2020-2026`, `Out [6]`):**
  - **1-Year Trend (`Simple Momentum 252d`)** is our strongest linear baseline (`0.77` **Gross Sharpe**, low turnover `30.8`).
  - **Pre-2020 Winner (`lookback=32`)** drops to `-0.07` **Sharpe** in 2020–2026 because a 32-day memory holds onto pre-crash states too long during sudden V-shaped reversals (COVID oil crash, 2022 Ukraine shock) [**memory lag at turning points**].
  - **Compact 20-Day LSTM (`lookback=20, hs=8, wd=1e-3`)** resets memory within one trading month → jumps to `1.27` **Gross Sharpe** (`17.6%` return).

Section 6: Seed Ensemble, Subperiods & 2×2 Ablation (`2:25 - 3:15` | 50 sec) — *Scroll to `Out [7]` & `Out [8]`*

- **5-Seed Ensemble (`Out [7]`):** Single seeds vary (`-0.33` to `1.27` [**initialization variance**]), so we average positions across 5 random seeds → cuts trading turnover in half (`64.8`), shrinks max loss to `-5.89%` [**Max Drawdown**], and yields `0.81` **Gross Sharpe**.
- **Training Windows & 2×2 Ablation (`Out [8]`):** Across 2013–2026 (`Table 6a`), the **3-year window (`0.68` SR)** beats noisy 2-year (`0.35`) and stale 5-year (`0.05`) windows (`0.88` full-sample for the Regime-Adaptive combination). And `Table 6b` (2×2 ablation) proves you need **both** multi-horizon features and the compact 20-day network (`1.27` together vs. `0.17` and `0.20` alone).

Part 2 — Tomàs (3:15 - 5:00 | 1 min 45 sec — Sections 7 & 8)

Section 7.1 & 7.2: Tuned Baselines & Transaction Costs (3:15 - 4:05 | 50 sec) — *Scroll to Out [9] & Out [10]*

- **Tuned Baselines (Out [9]):** Giving RidgeCV and 12-trial LightGBM the same 2012–2019 tuning budget still yields weak 2020–2026 performance (-0.44 and 0.15 Gross Sharpe).
- **Quadratic Trading Costs (Out [10]):** Costs scale with 60-day volatility and the square of trade size [compute_pnl_components] → fast-flipping models (1d Momentum, LightGBM, holding 1.5-3.3 days) lose **-3.2% to -6.1% per year**, wiping out their net edge (0.26 and -0.06 Net Sharpe).
- **Cost-Aware LSTM (WalkForwardMeanVarianceWithCost):** Passing yesterday's holdings [past_h] into the optimizer trades only partway toward the new target → cuts cost drag by **57%** (-0.36% vs. -0.84%) and achieves **1.30 Gross / 1.26 Net Sharpe** (and **1.23 Net Sharpe even at 3x costs**), while the **5-Seed Ensemble** achieves **0.78 Net Sharpe** (vs. 0.76 for 252d Momentum).

Section 7.3: Statistical Significance & Universe Stability (4:05 - 4:35 | 30 sec) — *Scroll to Out [11]*

- **HAC Significance (Out [11] Table):** Using autocorrelation-corrected standard errors [**Lo (2002) Newey-West HAC**], both the **Cost-Aware LSTM** (**1.26 Net SR,** $p = 0.0014$) and **5-Seed Ensemble** (**0.78 Net SR,** $p = 0.0327$) are statistically significant with 95% confidence intervals strictly above zero.
- **Leave-One-Out Test (Out [11] Bar Chart):** Dropping any single ETF one at a time keeps Gross Sharpe between **0.82 and 1.28** [**off-the-top universe stability**], proving the strategy doesn't depend on one lucky asset like oil (USO).

Section 8: Conclusion & Limitations (4:35 - 5:00 | 25 sec) — *Scroll to Section 8 Markdown*

- **Verdict:** A compact 20-day LSTM on multi-horizon momentum (5d-252d) with cost-aware optimization (1.26 Net SR) or 5-seed ensembling (0.78 Net SR, -5.89% Max DD) beats linear and tree baselines after costs.
- **4 Extensions:** (1) Train directly on portfolio Sharpe [**end-to-end loss**], (2) re-tune lookback every 1–2 years [**rolling nested HPO**], (3) add sideways-market filters (2016-2019), and (4) add futures curve slope features [**term-structure carry**].

2. Q&A Deep-Dive Reference (By Section — For Backup Only)

Q&A — Section 1: Data Pipeline & Stylized Facts

- **Why start in January 2007 (N = 5,158 business days)?**
  Commodity ETFs did not exist earlier (GLD launched Nov 2004, DBC/USO/SLV in 2006, DBA in Jan 2007). Starting at DBA's inception avoids missing-data imputation and covers 100% of the modern electronic ETF era.
- **Exact ETF Statistics (Out [1]):**
  - DBC (Broad): Mean 5.92%, Vol 19.36%, Sharpe 0.31, Skew -0.35, Kurtosis 3.46, Min -7.94%.
  - USO (Oil): Mean 2.53%, Vol 37.49%, Sharpe 0.07, Skew -0.55, Kurtosis 9.47, Min -25.32% (April 2020 WTI negative oil episode).
  - GLD (Gold): Mean 10.87%, Vol 18.14%, Sharpe 0.60, Skew -0.27, Kurtosis 7.17, Min -10.27%.
  - SLV (Silver): Mean 12.73%, Vol 33.00%, Sharpe 0.39, Skew -1.05, Kurtosis 13.41, Min -28.54%.
  - DBA (Agriculture): Mean 3.94%, Vol 16.68%, Sharpe 0.24, Skew -0.13, Kurtosis 6.41, Min -8.61%.
- **Doesn't DBC overlap with USO, GLD, and DBA?**
  Yes, DBC is a broad composite commodity index ETF (corr = 0.79 with USO). However, the Markowitz covariance matrix V in MeanVariance explicitly accounts for cross-asset correlations, and our leave-one-out test (Excl. DBC) still achieves a Gross Sharpe above 1.0.
- **How is target volatility scaled?**
  Daily target risk is TARGET_VOL = 0.10 / sqrt(260) = 0.006202 (10% annualized). Realized OOS volatility can deviate slightly (12-14% for single models, 6.03% for the ensemble) because covariance is estimated on the trailing window and applied out-of-sample.

Q&A — Sections 2 & 3: Features, Baselines & PyTorch LSTM Wrapper

- **How do you prevent look-ahead leakage in Backtester and PyTorchLSTMRegressor?**
  1. Target at index t is ret.shift(-1) (r_{t+1}), and PnL is h.shift(1).mul(ret).
  2. StandardScaler(with_mean=False) is inside make_pipeline(...), so scaling parameters are fit strictly inside each 756-day training fold.
  3. Validation split (val_split=0.15, ~110 days) is sliced chronologically from the end of each training fold *before* mini-batch shuffling.
  4. Sequence boundary buffer (last_train_X_): because test_size=21 is shorter than or comparable to lookback (20-60 days), fit() caches the last lookback rows of X_train and prepends them in predict(X_test) so test day 1 uses [t - L + 1, ..., t] without peeking ahead or dropping initial test days.

- **Why initialize the forget-gate bias to $b_f = +1.0$?**
  In PyTorch's `nn.LSTM`, the bias vector has length $4 \times$ hidden_size ordered `[input, forget, cell, output]`. Setting the second quarter (`n//4 : n//2`) to `+1.0` ensures $\sigma(b_f) \approx 0.73$ at epoch 1 so gradients flow across the 20-step sequence instead of vanishing immediately.
- **Why Huber loss ($\delta = 0.01$) instead of MSE?**
  A 1% daily move ($\delta = 0.01$) is roughly 1 standard deviation for a diversified commodity basket. For normal days ($|e| \leq 0.01$), Huber is quadratic ($0.5e^2$); for extreme tail shocks ($|e| > 0.01$, like `-25%` in `USO`), gradients are capped at $\pm\delta$, preventing a single outlier day from dominating AdamW updates.

Q&A — Sections 4 & 5: Hyperparameter Search & Out-of-Sample Results

- **Exact Random Search Space (`2012-2019`, 15 trials, `random_state=42`):**
  `hidden_size` $\in$ randint$(6, 17)$, `lookback` $\in$ randint$(15, 45)$, `weight_decay` $\in$ loguniform$(10^{-4}, 5 \times 10^{-3})$, `dropout` $\in$ uniform$(0.10, 0.35)$, `lr` $\in$ loguniform$(5 \times 10^{-4}, 2 \times 10^{-3})$.
- **Wasn't choosing `Compact LSTM (lb=20, hs=8, wd=1e-3)` after seeing 2020–2026 a form of hindsight?**
  We explicitly report **both** the frozen 2012–2019 random search winner (`Trial 9: lb=32, hs=9, wd=3.4e-4`, OOS SR `-0.07`) and the canonical 1-month (`lb=20`) strongly regularized architecture (`wd=1e-3`, OOS SR `1.27`). Showing both side-by-side demonstrates *why* static pre-2020 hyperparameter selection fails across structural breaks (memory lag) and motivates rolling nested HPO in Section 8.

Q&A — Section 6: Multi-Seed Ensembling & Subperiod Sensitivity

- **Why is the 5-Seed Ensemble Sharpe (`0.81`) higher than the average of the 5 seeds (`0.36`)?**
  The 5 seeds (`42: 1.27`, `101: 0.35`, `2024: -0.17`, `7: 0.69`, `999: -0.33`) have mean return `4.78%` (the exact average of the 5 seeds' returns), **but** because their forecast errors are weakly correlated across seeds, averaging their holdings reduces portfolio volatility from `~14.1%` down to **`6.03%`**. Since Sharpe $= \mu/\sigma$, the 57% drop in denominator volatility boosts the ensemble Sharpe to `0.809` (`0.81`) and cuts max net drawdown to `-5.89%`.
- **Exact Subperiod Sharpe Table (`Table 6a`, 2013–2026):**
  - **2013–2015 (Post-GFC Bear Crash):** `252d Mom = 0.63`, `Pre-2020 LSTM (lb=32) = 0.76`, `Compact LSTM (3y) = 0.22`.
  - **2016–2019 (Sideways Recovery):** `252d Mom = 0.26`, `Pre-2020 LSTM (lb=32) = 0.05`, `Compact LSTM (3y) = -0.41`.
  - **2020–2026 (Pandemic & Inflation):** `252d Mom = 0.77`, `Pre-2020 LSTM (lb=32) = -0.07`, `Compact LSTM (2y) = 0.99`, `Compact LSTM (3y) = 1.27`, `Compact LSTM (5y) = 0.34`.
  - **Full 2013–2026 (13 Years):** `252d Mom = 0.61`, `Compact LSTM (2y) = 0.35`, `Compact LSTM (3y) = 0.68`, `Compact LSTM (5y) = 0.05`, `Regime-Adaptive LSTM = 0.88`.

Q&A — Section 7: Baselines, Transaction Costs & Statistical Validation

- **How were `RidgeCV` and `MultiLGBMRegressor` tuned?**
  - `RidgeCV`: Tuned across `alphas = [0.01, 0.1, 1, 10, 100, 1000]` inside every walk-forward fold (`mode alpha = 1000.0`; IS SR `-0.062`, OOS Gross SR `-0.44`, Net SR `-0.46`).
  - `MultiLGBMRegressor`: 12-trial `ParameterSampler` on 2012–2019 (`n_estimators=30`, `min_child_samples=10`, `lr=0.0117`, `max_depth=5`; IS SR `0.142`, OOS Gross SR `0.15`, Net SR `-0.06`).
- **How is the volatility-liquidity cost factor (`vlf`) defined?**
  $\text{vlf}_{t,i} = \frac{\sigma_{t,i}^{(60\text{d})}}{1/N} \times 2 \times 10^{-3}$ (with $N = 5$ ETFs), computed strictly point-in-time using trailing 60-day volatility (expanding volatility for the first 20 days, zero backward filling). Daily impact cost is $\sum_{i=1}^{5} \text{vlf}_{t,i}(\Delta h_{t,i})^2$.
- **Why did you subclass `MeanVarianceWithCost` and `BacktesterWithCost`?**
  1. `skfin`'s default `BacktesterWithCost` assumes `X.shape[1] == y.shape[1]`, whereas `X_mom` has 25 features and `y_mom` has 5 assets (`MultiAssetBacktesterWithCost` uses `y.columns` for holdings).
  2. `WalkForwardMeanVarianceWithCost` averages `vol_liquidity_factor` across the 21-day test block, takes the most recent holding `past_h[-1:]`, computes Gârleanu–Pedersen cost-aware holdings, and rescales ex-ante portfolio volatility $\sqrt{h^\top V h}$ to `TARGET_VOL`.
- **How does the Lo (2002) Newey-West HAC test work (`Out [11]`)?**
  Standard IID Sharpe standard error is $\sqrt{(1 + 0.5\,\text{SR}_d^2)/T}$. To account for serial correlation and volatility clustering, we compute the influence function of the Sharpe ratio $\phi_t = \frac{r_t - \mu}{\sigma} - \frac{\mu}{2\sigma^3}\left((r_t - \mu)^2 - \sigma^2\right)$ and apply a Bartlett kernel up to lag $q = \lfloor 4(T/100)^{2/9} \rfloor = 7$ days (`T = 1,770` OOS days).
- **Exact Leave-One-ETF-Out OOS Gross Sharpe Ratios (`Out [11]` Bar Chart):**
  Full 5-ETF universe: `1.27` | Excluding any single ETF stays strictly between **`0.82` and `1.28`**.